PRACTICAL 2

Semantic HTML5 Pages with Accessibility-Ready Structure

1. Problem Definition

StudentHub is a semester-long student portal project. In Practical 2, static HTML5 page skeletons are developed for the main StudentHub modules. The pages use semantic HTML5 elements, accessible forms, meaningful headings, alternative text for images, and consistent navigation.

The objective is to create clean, linked, and accessibility-ready HTML page structures that can later be styled and enhanced using CSS and JavaScript.

2. Pages Developed

1. Home
2. About
3. Register
4. Login
5. Dashboard
6. Events
7. Profile
8. Contact
9. Admin
10. FAQ
11. Feedback

3. Common Semantic Page Structure

Each StudentHub page follows a consistent HTML5 semantic structure:

```
<header>
  <nav>
    Navigation links
  </nav>
</header>

<main>
  <section>
    Main page content
  </section>

  <article>
    Independent content
  </article>
```

```
    <aside>
        Supporting information
    </aside>
</main>

<footer>
    Footer information
</footer>
```

4. Semantic HTML5 Elements Used

Element	Purpose
<header>	Contains the introductory content and website branding.
<nav>	Contains the main navigation links.
<main>	Contains the primary content of the page.
<section>	Groups related content into meaningful sections.
<article>	Represents independent or self-contained content.
<aside>	Contains supporting or related information.
<footer>	Contains footer information and secondary links.
<h1> to <h6>	Provides a logical content heading hierarchy.
<form>	Contains user input controls.
<label>	Provides an accessible name for form controls.
<button>	Provides an interactive action.
<img>	Displays images with alternative text using alt.

5. Common Navigation

The same navigation structure is maintained across all StudentHub pages to provide consistent navigation.

Home
About
Register
Login
Dashboard
Events
Profile
Contact
FAQ
Feedback
Admin

6. HTML Page Structures

6.1 Home Page

The Home page introduces the StudentHub portal and provides links to important sections.

```
<header>
  <h1>StudentHub</h1>
  <nav>
    Home | About | Register | Login
  </nav>
</header>

<main>
  <section>
    <h2>StudentHub Portal</h2>
    <p>Your Academic Information Hub</p>
  </section>

  <section>
    <h2>Features</h2>

    <article>
      <h3>Academic Information</h3>
      <p>Access academic information.</p>
    </article>

    <article>
      <h3>College Events</h3>
      <p>View upcoming college events.</p>
    </article>

    <article>
      <h3>Student Profile</h3>
      <p>Manage student profile information.</p>
    </article>
  </section>
</main>

<footer>
  <p>StudentHub © 2026</p>
</footer>
```

6.2 About Page

The About page provides information about the StudentHub portal.

```
<header>
```

```
  <h1>StudentHub</h1>
  <nav>Home | About | Register | Login</nav>
</header>

<main>
  <section>
    <h2>About StudentHub</h2>
    <p>
      StudentHub is a centralized student portal
      for academic and college information.
    </p>
  </section>

  <section>
    <h2>Our Purpose</h2>
    <article>
      <h3>Centralized Information</h3>
      <p>Important information is available in one place.</p>
    </article>
    <article>
      <h3>Easy Access</h3>
      <p>Students can access information easily.</p>
    </article>
  </section>
</main>

<footer>
  <p>StudentHub © 2026</p>
</footer>
```

6.3 Register Page

The Register page contains an accessible registration form.

```
<header>
  <h1>StudentHub</h1>
  <nav>Home | About | Login</nav>
</header>

<main>
  <section>
    <h2>Create Student Account</h2>

    <form>
      <label for="fullName">Full Name:</label>
      <input type="text" id="fullName" name="fullName" required>
```

```
        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required>

        <label for="studentId">Student ID:</label>
        <input type="text" id="studentId" name="studentId" required>

        <label for="password">Password:</label>
        <input type="password" id="password" name="password" required>

        <label for="course">Course:</label>
        <select id="course" name="course">
          <option value="">Select Course</option>
          <option value="aiml">AIML</option>
          <option value="ce">Computer Engineering</option>
          <option value="it">Information Technology</option>
        </select>

        <button type="submit">Register</button>
      </form>
    </section>
</main>

<footer>
    <p>StudentHub © 2026</p>
</footer>
```

6.4 Login Page

The Login page allows registered users to access StudentHub.

```
<header>
    <h1>StudentHub</h1>
    <nav>Home | About | Register | Login</nav>
</header>

<main>
    <section>
      <h2>Login</h2>

      <form>
        <label for="loginEmail">Email:</label>
        <input type="email" id="loginEmail" name="loginEmail" required>

        <label for="loginPassword">Password:</label>
        <input type="password" id="loginPassword" name="loginPassword" required>

        <button type="submit">Login</button>
```

```
      </form>
   </section>
</main>

<footer>
   <p>StudentHub © 2026</p>
</footer>
```

6.5 Dashboard Page

The Dashboard page provides a summary of student information.

```
<header>
   <h1>StudentHub</h1>
   <nav>Dashboard | Profile | Courses | Attendance | Events | Logout</nav>
</header>

<main>
   <section>
      <h2>Student Dashboard</h2>

      <article>
         <h3>Courses</h3>
         <p>6 active courses</p>
      </article>

      <article>
         <h3>Attendance</h3>
         <p>85% overall attendance</p>
      </article>

      <article>
         <h3>Assignments</h3>
         <p>3 pending assignments</p>
      </article>

      <aside>
         <h3>Important Notice</h3>
         <p>Check upcoming college events.</p>
      </aside>
   </section>
</main>

<footer>
   <p>StudentHub © 2026</p>
</footer>
```

6.6 Events Page

The Events page displays upcoming college events.

```
<header>
  <h1>StudentHub</h1>
  <nav>Home | Dashboard | Events | Profile</nav>
</header>

<main>
  <section>
    <h2>College Events</h2>

    <article>
      <h3>Technical Workshop</h3>
      <p>Date: 15 August 2026</p>
      <p>Modern web development workshop.</p>
    </article>

    <article>
      <h3>Sports Day</h3>
      <p>Date: 25 August 2026</p>
      <p>Annual college sports activities.</p>
    </article>

    <article>
      <h3>Student Orientation</h3>
      <p>Date: 1 September 2026</p>
      <p>Student orientation programme.</p>
    </article>
  </section>
</main>

<footer>
  <p>StudentHub © 2026</p>
</footer>
```

6.7 Profile Page

The Profile page displays student information.

```
<header>
  <h1>StudentHub</h1>
  <nav>Dashboard | Profile | Courses | Logout</nav>
</header>

<main>
  <section>
```

```
      <h2>Student Profile</h2>

      <img src="../images/profile.jpg"
          alt="Student profile photograph">

      <p><strong>Name:</strong> Student Name</p>
      <p><strong>Student ID:</strong> SH101</p>
      <p><strong>Email:</strong> student@example.com</p>
      <p><strong>Course:</strong> AIML</p>
   </section>
</main>

<footer>
   <p>StudentHub © 2026</p>
</footer>
```

6.8 Contact Page

The Contact page contains contact information and an accessible contact form.

```
<header>
   <h1>StudentHub</h1>
   <nav>Home | About | Contact | Login</nav>
</header>

<main>
   <section>
      <h2>Contact StudentHub</h2>

      <address>
         Email: studenthub@example.com<br>
         Phone: +91 XXXXX XXXXX
      </address>
   </section>

   <section>
      <h2>Contact Form</h2>

      <form>
         <label for="contactName">Name:</label>
         <input type="text" id="contactName" name="contactName" required>

         <label for="contactEmail">Email:</label>
         <input type="email" id="contactEmail" name="contactEmail" required>

         <label for="message">Message:</label>
         <textarea id="message" name="message" rows="5"></textarea>
```

```
        <button type="submit">Send Message</button>
      </form>
    </section>
</main>

<footer>
    <p>StudentHub © 2026</p>
</footer>
```

6.9 Admin Page

The Admin page provides administrative management options.

```
<header>
    <h1>StudentHub Admin</h1>
    <nav>Dashboard | Students | Faculty | Courses | Events | Logout</nav>
</header>

<main>
    <section>
      <h2>Admin Dashboard</h2>

      <article>
        <h3>Manage Students</h3>
        <p>View and manage student records.</p>
      </article>

      <article>
        <h3>Manage Faculty</h3>
        <p>View and manage faculty records.</p>
      </article>

      <article>
        <h3>Manage Courses</h3>
        <p>Manage course information.</p>
      </article>
    </section>
</main>

<footer>
    <p>StudentHub © 2026</p>
</footer>
```

6.10 FAQ Page

The FAQ page contains frequently asked questions.

```
<header>
```

```
  <h1>StudentHub</h1>
  <nav>Home | Dashboard | FAQ | Contact</nav>
</header>

<main>
  <section>
    <h2>Frequently Asked Questions</h2>

    <article>
      <h3>What is StudentHub?</h3>
      <p>StudentHub is a centralized student portal.</p>
    </article>

    <article>
      <h3>How can I register?</h3>
      <p>Use the Register page to create an account.</p>
    </article>

    <article>
      <h3>How can I access my dashboard?</h3>
      <p>Login using your registered credentials.</p>
    </article>
  </section>
</main>

<footer>
  <p>StudentHub © 2026</p>
</footer>
```

6.11 Feedback Page

The Feedback page allows users to submit feedback.

```
<header>
  <h1>StudentHub</h1>
  <nav>Home | Dashboard | Feedback | Contact</nav>
</header>

<main>
  <section>
    <h2>Student Feedback</h2>

    <form>
      <label for="feedbackName">Name:</label>
      <input type="text" id="feedbackName" name="feedbackName" required>

      <label for="feedbackEmail">Email:</label>
```

```
        <input type="email" id="feedbackEmail" name="feedbackEmail" required>

        <label for="rating">Rating:</label>
        <select id="rating" name="rating">
          <option value="5">Excellent</option>
          <option value="4">Very Good</option>
          <option value="3">Good</option>
          <option value="2">Average</option>
          <option value="1">Poor</option>
        </select>

        <label for="feedback">Feedback:</label>
        <textarea id="feedback" name="feedback" rows="5"></textarea>

        <button type="submit">Submit Feedback</button>
      </form>
    </section>
</main>

<footer>
    <p>StudentHub © 2026</p>
</footer>
```

7. Accessibility Checklist

Accessibility Requirement	Implementation	Status
Semantic HTML5 tags	header, nav, main, section, article, aside and footer are used.	Yes
Heading hierarchy	Headings are arranged logically using h1, h2 and h3.	Yes
Form labels	Every important form control has a corresponding label.	Yes
Image alt text	Images include meaningful alt attributes.	Yes
Navigation links	Navigation links are available across pages.	Yes
Keyboard accessibility	Native links, buttons and form controls can be accessed using keyboard.	Yes
Required fields	Important form fields use required attributes.	Yes
Readable content	Clear headings and meaningful text are used.	Yes
Consistent structure	Common header, navigation and footer are used.	Yes

8. Key Questions / Analysis / Interpretation

Q1. Are semantic tags such as header, nav, main, section, article, aside, and footer used properly?

Yes. Semantic HTML5 elements are used according to their purpose. The header contains branding and introductory content, nav contains navigation links, main contains the primary content, section groups related information, article contains independent content, aside contains supporting information, and footer contains footer information.

Q2. Are form labels, headings, image alt text, and navigation links accessible?

Yes. Form controls are connected to labels using the for and id attributes. Headings are organized in a logical hierarchy. Images use descriptive alt text. Navigation is provided using standard anchor elements so it can be accessed using keyboard and assistive technologies.

Q3. Is the page structure consistent across all pages?

Yes. All pages follow a common structure consisting of header, navigation, main content, and footer. This provides a consistent user experience and makes the website easier to understand and maintain.

9. Project Folder Structure

```
StudentHub/
│
├── index.html
│
├── pages/
│   ├── about.html
│   ├── register.html
│   ├── login.html
│   ├── dashboard.html
│   ├── events.html
│   ├── profile.html
│   ├── contact.html
│   ├── admin.html
│   ├── faq.html
│   └── feedback.html
│
├── css/
│   └── style.css
│
├── js/
│   └── script.js
│
└── images/
    └── profile.jpg
```

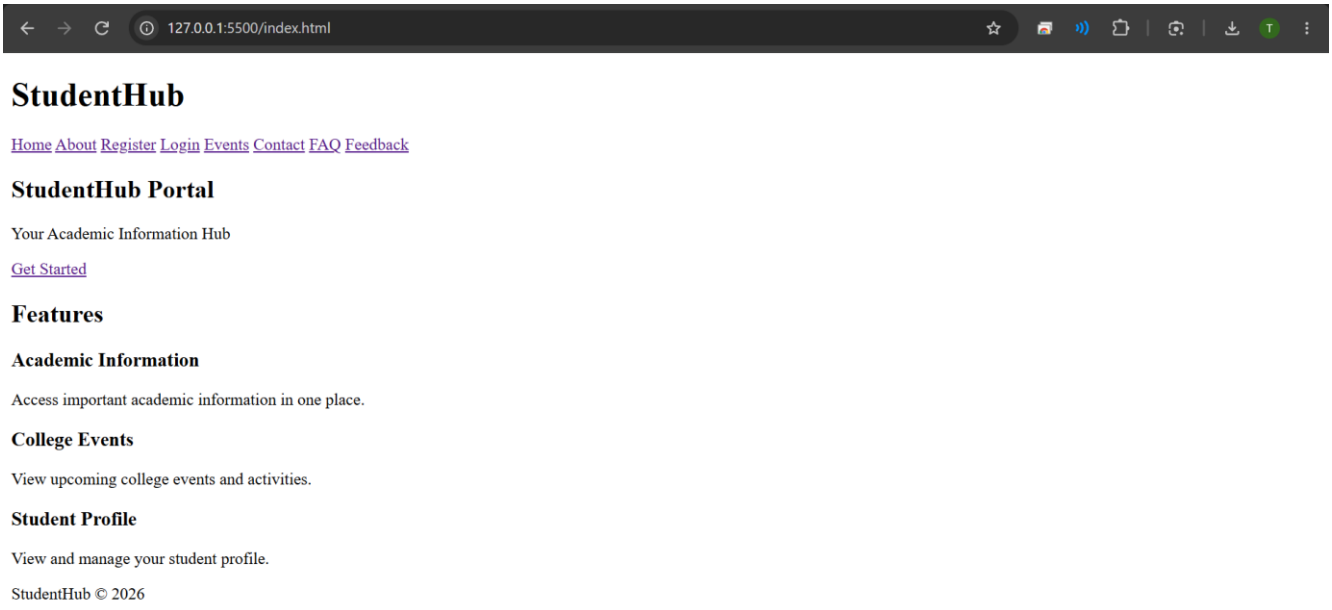
10. Submission

- Static HTML page screenshots
- HTML source code
- Accessibility checklist
- Linked StudentHub pages
- Screenshots showing the completed pages

11. Conclusion

The StudentHub static HTML5 pages were successfully structured using semantic HTML5 elements and accessibility-ready practices. The pages use consistent navigation, meaningful headings, accessible forms, alternative text for images, and a common page structure. This provides a strong foundation for the responsive CSS implementation in the next practical.

12. Screenshots





StudentHub

Home About Register Login Events Contact FAQ Feedback

Create Student Account

Full Name:

Email:

Student ID:

Password:

Confirm Password:

Course: Select Course

Register

StudentHub © 2026



StudentHub

Home About Register Login Events Contact FAQ Feedback

Login

Email:

Password:

Login

Don't have an account? Register

StudentHub © 2026



StudentHub

Home About Dashboard Events Contact FAQ Feedback

Contact StudentHub

Email: studenthub@example.com
Phone: +91 00000 00000

Contact Form

Name:

Email:

Message:

Send Message

StudentHub © 2026



StudentHub

Home Dashboard Events Contact FAQ Feedback

Student Feedback

Name:

Email:

Rating: Select Rating

Feedback:

Submit Feedback

StudentHub © 2026